

Ephaptic A β -to-C-Fiber Crosstalk in Pathologically Restricted Nerve Clefts: A Closed-Loop Core-Conductor Study of Spatial and Temporal Summation

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Abstract: Dynamic mechanical allodynia is hypothesized to involve non-synaptic (“ephaptic”) cross-excitation between myelinated A β mechanoreceptive afferents and unmyelinated C-fiber nociceptors at sites of focal demyelination or nerve compression. We present a closed-loop, one-dimensional core-conductor model coupling a Chiu-Ritchie-Rogart-Stagg-Sweeney (CRRSS) A β axon to a C-fiber through a shared, restricted extracellular cleft, using algebraic elimination of capacitive currents. C-fibers used classical Hodgkin-Huxley kinetics (Phase 1) or literature-derived Nav1.8/Nav1.9 kinetics (Phase 2, robustness check). Single-fiber-pair crosstalk remained subthreshold under single-pulse stimulation in both models (≤ 1.325 mV, Phase 1; ≤ 1.76 mV, Phase 2) across cleft widths of 20 nm to 5 μ m. In a bundle of $n \in [1, 50]$ synchronized A β fibers, return currents produced saturating subthreshold depolarization within 5 ms (+6.2 mV, Phase 1; +10.4 mV, Phase 2, at $n=50$), limited by axoplasmic shunting; a 1.5-ms onset-dispersion window attenuated it by 92.9–93.8%. In the ungrounded full-length model, a 100-Hz train produced a propagated C-fiber action potential (≈ 0.41 m/s) during the second pulse interval, but it was initiated by the first pulse at the sealed stimulation boundary and delayed by slow conduction; a single pulse yields the same spike, so it does not demonstrate temporal summation. In a grounded interior-lesion control, endogenous crosstalk evoked no action potentials in either phase, including under 800-Hz trains. Isolated A β conduction velocity (41.03 m/s) and event classifications were robust to spatial refinement, though the single-pulse $n=25$ peak amplitude was not fully timestep-converged. Here, ephaptic coupling produced subthreshold depolarization but no threshold crossing without a boundary-initiated spike.

Keywords: ephaptic coupling, dynamic mechanical allodynia, core-conductor model, Nav1.8/Nav1.9 kinetics, spatial summation, temporal summation

1 Introduction

Dynamic mechanical allodynia—pain evoked by light, moving tactile stimuli such as light brushing or the contact of clothing—is a debilitating hallmark of peripheral neuropathic pain conditions, including painful diabetic neuropathy (Asmedi et al., 2018), post-herpetic neuralgia, and nerve entrapment syndromes such as trigeminal neuralgia (Rappaport & Devor, 1994). A prominent biophysical hypothesis holds that at sites of focal demyelination, chronic mechanical compression, or traumatic nerve injury, the normally independent electrical activity of large myelinated A β mechanoreceptive afferents and small unmyelinated C-fiber nociceptors can become directly coupled through their shared, pathologically restricted extracellular space. This non-synaptic electrical interaction, termed “ephaptic” coupling, was first characterized biophysically by Arvanitaki (1942) and by Katz and Schmitt (1940). Experimental evidence for cross-excitation between afferent classes has since been documented in mammalian dorsal root ganglia (Amir & Devor, 2000; Devor & Wall, 1990) and in chronically injured peripheral nerve trunks (Rasminsky, 1980; Seltzer & Devor, 1979). Under this hypothesis, low-threshold tactile activity conducted by A β fibers could directly evoke ectopic action potentials in neighboring nociceptive axons,

providing a peripheral mechanism for tactile allodynia that operates upstream of, or in concert with, central sensitization.

Computational investigations of ephaptic coupling in nervous tissue have traditionally followed two distinct approaches: open-loop forward models, in which the extracellular field generated by an active fiber is imposed as an external driving potential on an adjacent passive or excitable cable without reciprocal feedback (Goldwyn & Rinzel, 2016; Lindén et al., 2014), and closed-loop formulations restricted to homogeneous fiber populations, such as bundles of myelinated axons or central white-matter tracts (Reutskiy et al., 2003; Schmidt et al., 2021; Schmidt & Knösche, 2022). Foundational analytical and numerical work by Reutskiy, Rossoni, and Tirozzi (2003) demonstrated that bundle geometry and inter-fiber packing profoundly modulate conduction velocity and recruitment thresholds in demyelinated nerve. More recently, generalized higher-dimensional formulations of interacting parallel axon networks have elucidated the complex dynamical dependencies of ephaptic crosstalk on spatial packing, boundary conditions, and coupling topologies (Chawla & Morgera, 2024; Chawla et al., 2021). Fully coupled Extracellular–Membrane–Intracellular (EMI) finite-element formulations have further advanced the field by resolving three-dimensional micro-domains around excitable cells (Jæger & Tveito, 2022, 2026). Nevertheless, closed-loop bidirectional coupling between physiologically distinct fiber classes—specifically a fast myelinated A β mechanoreceptor and an unmyelinated C-fiber nociceptor—remains incompletely characterized, particularly concerning whether multi-fiber spatial convergence or repetitive temporal trains can bridge the threshold gap where single-pair coupling fails.

This paper reports a closed-loop core-conductor model developed to evaluate this interaction systematically across two complementary modeling phases. Phase 1 establishes a baseline system coupling a Chiu-Ritchie-Rogart-Stagg-Sweeney (CRRSS) myelinated A β axon to a classical Hodgkin-Huxley (HH) C-fiber through a shared extracellular cleft. Using this baseline, we test whether single-pair crosstalk can trigger ectopic nociceptor spikes under single-pulse stimulation, tonic peripheral sensitization, or high-frequency temporal summation. Recognizing that classical HH kinetics lack the tetrodotoxin-resistant (TTX-R) conductances characteristic of mammalian nociceptors, Phase 2 replaces the classical Na⁺ current with literature-derived Nav1.8 and Nav1.9 kinetics (Akopian et al., 1996; Cummins et al., 1999; Klein et al., 2017). Phase 2 is designed not as an independent biological validation of Phase 1, but as a biological sensitivity and robustness check within the same one-dimensional core-conductor framework. We then extend Phase 2 to a multi-fiber bundle configuration in which an ensemble of synchronized or temporally dispersed A β fibers converges upon a shared extracellular compartment with one C-fiber.

We report that single-fiber-pair crosstalk remains strictly subthreshold under single-pulse stimulation across both channel models. In contrast, multi-fiber spatial summation produces substantial subthreshold depolarizations that asymptotically saturate due to parallel axoplasmic conductance shunting. Furthermore, in the ungrounded full-length model a 100-Hz train yields a propagated C-fiber action potential during the second pulse interval; because a single pulse produces the same spike when observed for long enough, we attribute it to boundary initiation and slow conduction rather than to demonstrated temporal summation, and a grounded interior-lesion control evoked no action potentials.

2 Methods

2.1 Coupled Core-Conductor Cable Equations

We consider two parallel axons of length $L = 10$ mm embedded in a shared, longitudinally restricted extracellular cleft with resistance per unit length r_e (Ω/m). Axon 1 ($k=1$) represents the myelinated A β fiber, and axon 2 ($k=2$) represents the unmyelinated C-fiber. For each axon, the transmembrane potential is defined as $v_{m,k}(z,t) = u_{i,k}(z,t) - u_e(z,t)$, where $u_{i,k}$ is the intracellular potential and u_e is the shared extracellular cleft potential. The coupled core-conductor cable equations are given by:

$$\frac{\partial v_{m,k}}{\partial t} = \frac{1}{C_{m,k}} \left[\frac{d_{axon,k}}{4\rho_i} \left(\frac{\partial^2 v_{m,k}}{\partial z^2} + \frac{\partial^2 u_e}{\partial z^2} \right) - I_{ion,k} + I_{stim,k} \right]$$

where $d_{axon,k}$ is the inner axonal core diameter ($d_{axon,1} = 0.7 D_1 = 7.0 \mu m$ for the myelinated A β fiber with outer diameter $D_1 = 10 \mu m$, representing an established g -ratio of 0.7; and $d_{axon,2} = d_2 = 1.0 \mu m$ for the unmyelinated C-fiber). The parameter $\rho_i = 54.7 \Omega \cdot cm = 0.547 \Omega \cdot m$ is the axoplasmic resistivity (Sweeney et al., 1987), $C_{m,k}$ is the specific membrane capacitance (F/m^2), $I_{ion,k}$ is the total transmembrane ionic current density (A/m^2), and $I_{stim,k}$ is the applied intracellular stimulation current density (A/m^2). All current densities and capacitances are defined per unit inner-axonal membrane area ($\pi d_{axon,k}$), ensuring mathematical consistency with the core-conductor derivation of the axial diffusion term. Symbols, parameters, and SI units are defined in Table 3. Sealed-end (Neumann) boundary conditions are enforced at both cable ends ($z = 0$ and $z = L$):

$$\left. \frac{\partial v_{m,k}}{\partial z} \right|_{z=0,L} = 0, \quad \left. \frac{\partial u_e}{\partial z} \right|_{z=0,L} = 0$$

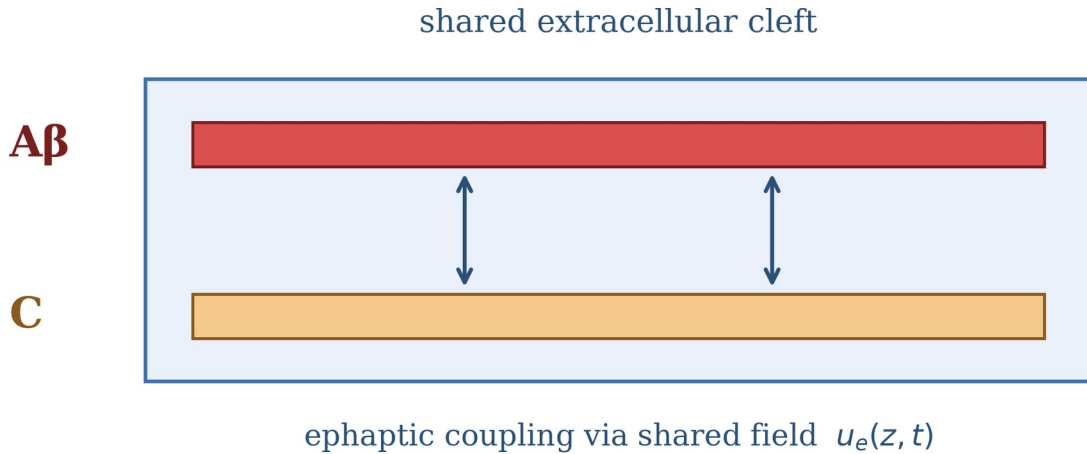


Fig. 1 Model architecture (schematic). A myelinated A β axon (CRRSS kinetics, discrete nodes of Ranvier) and an unmyelinated C-fiber (classical HH or Nav1.8/Nav1.9 kinetics) are embedded in a pathologically restricted extracellular cleft of width w_{cleft} . Both fibers' transmembrane currents drive a regularized Poisson equation for the shared extracellular potential $u_e(z,t)$, which is solved at each time step and fed back into both cable equations as an ephaptic driving term (double-headed arrows), closing the loop

2.2 Membrane Kinetics and Conduction Properties

2.2.1 Myelinated A β Fiber (CRRSS Kinetics)

The A β axon employs the Chiu-Ritchie-Rogart-Stagg-Sweeney (CRRSS) nodal kinetics (Chiu et al., 1979; Sweeney et al., 1987). Active ionic currents are localized to discrete physical nodes of Ranvier of fixed physical length $l_{node} = 1.0 \mu\text{m}$, spaced at internodal intervals of $\Delta L = 1.0 \text{ mm}$ along the 10 mm cable (10 nodes total, at $z = 0, 1, \dots, 9 \text{ mm}$). To ensure that spatial grid refinement (Δz) does not alter the physical membrane area of the node, we apply a consistent homogenization scheme: for any computational compartment of width Δz , the fraction occupied by an active node is $f_{node} = l_{node} / \Delta z$ ($f_{node} \leq 1$). The effective compartment capacitance and leak conductance are weighted accordingly:

$$C_{m,1} = f_{node} C_{m,node} + (1 - f_{node}) C_{m,internode}$$

$$g_{leak,1} = f_{node} g_{leak,node} + (1 - f_{node}) g_{leak,internode}$$

Physical nodal parameters are: $\bar{g}_{Na} = 1445 \text{ mS/cm}^2 = 14450 \text{ S/m}^2$, $g_{leak,node} = 128 \text{ mS/cm}^2 = 1280 \text{ S/m}^2$, $C_{m,node} = 2.0 \mu\text{F/cm}^2$, $E_{Na} = +35.64 \text{ mV}$, and $E_{leak,1} = -80.0 \text{ mV}$ (Sweeney et al., 1987). Internodal compact myelin is parameterized with $C_{m,internode} = 0.005 \mu\text{F/cm}^2$ and $g_{leak,internode} = 0.006 \text{ mS/cm}^2$, with active Na⁺ channels absent.

The gating rate equations follow the mammalian CRRSS formulation (Chiu et al., 1979; Sweeney et al., 1987) with corrected monotonic h -gate inactivation designations:

$$\alpha_m(v) = \frac{(97.0 + 0.363(v_{mV} + 80.0)) \times 10^3}{1.0 + \exp(-(v_{mV} + 49.0)/5.3)} \text{ s}^{-1}$$

$$\beta_m(v) = \alpha_m(v) \exp(-(v_{mV} + 56.2)/4.17)$$

$$\beta_h(v) = \frac{15.6 \times 10^3}{1.0 + \exp(-(v_{mV} + 56.0)/10.0)} \text{ s}^{-1}$$

$$\alpha_h(v) = \beta_h(v) \exp(-(v_{mV} + 74.5)/5.0)$$

where $v_{mV} = v_{m,1} \times 10^3$, corresponding to the classic deviation from resting potential $V = v_{mV} + 80.0 \text{ mV}$ in Chiu et al. (1979). The inactivation curve $h_\infty(v) = \alpha_h / (\alpha_h + \beta_h)$ decreases monotonically from ≈ 1.0 at hyperpolarized potentials to 0.0 upon depolarization, with half-inactivation at $V_{1/2} \approx -74.5 \text{ mV}$. Rates are scaled to 37 °C using $Q_{10} = 3.0$ (Sweeney et al., 1987). Under this parameterization, an isolated A β fiber stimulated at node 0 (2 nA, 0.2 ms) generates a brief, physiological saltatory action potential, reaching an absolute peak of approximately +0.44 mV at $z = 4 \text{ mm}$ from a resting potential of -80.0 mV, corresponding to an action-potential excursion of 80.44 mV, conducting at **41.03 m/s** (measured via maximum upstroke dV/dt between 4.0 mm and 8.0 mm), well within the established mammalian range (30–60 m/s). The coupled simulations of Section 3.2 onward use a stronger 100 nA stimulus, which produces a large local transient at the stimulated end (Section 3.5.1); the 2 nA drive is used for all uncoupled A β validation. Steady-state gating curves and time constants are shown in Fig. 2.

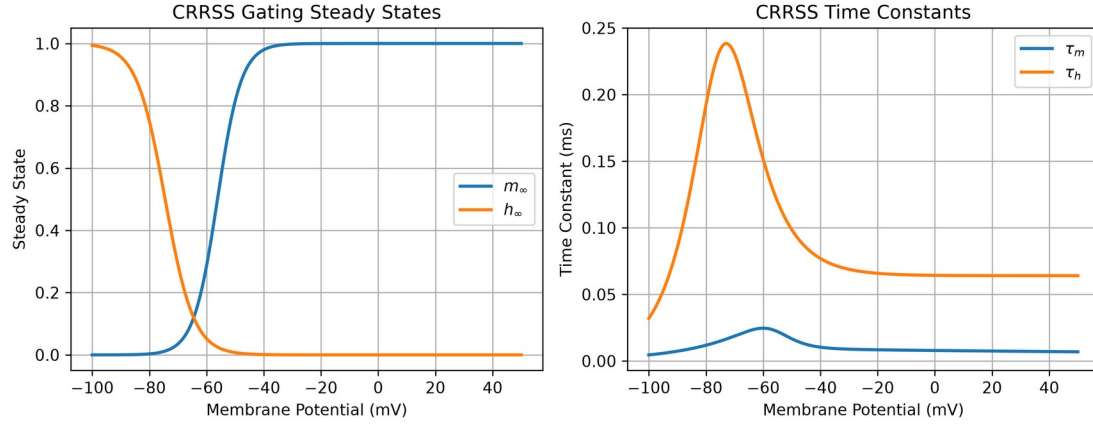


Fig. 2 CRRSS nodal gating kinetics used for the myelinated A β fiber. Left, steady-state activation (m_∞) and inactivation (h_∞) as functions of membrane potential; right, corresponding time constants τ_m and τ_h . The inactivation curve decreases monotonically with depolarization, confirming the corrected h-gate rate equations

2.2.2 C-Fiber Membrane Models

The unmyelinated C-fiber ($d_2 = 1.0 \mu\text{m}$, $C_{m,2} = 1.0 \mu\text{F}/\text{cm}^2$) is modeled continuously along its length. In Phase 1, it follows classical Hodgkin-Huxley kinetics (Hodgkin & Huxley, 1952): $\bar{g}_{Na} = 120 \text{ mS}/\text{cm}^2$, $\bar{g}_K = 36 \text{ mS}/\text{cm}^2$, $g_{leak} = 0.3 \text{ mS}/\text{cm}^2$, $E_{Na} = +50.0 \text{ mV}$, $E_K = -77.0 \text{ mV}$, and $E_{leak,2} = -54.4 \text{ mV}$, resting stably at -65.0 mV . The delayed-rectifier K^+ conductance obeys $I_K = \bar{g}_K n^4 (v_2 - E_K)$ with gating kinetics:

$$\frac{dn}{dt} = \alpha_n(v)(1 - n) - \beta_n(v)n$$

$$\alpha_n(v) = \frac{0.01(v_{mV} + 55.0)}{1.0 - \exp(-(v_{mV} + 55.0)/10.0)} \times 10^3 \text{ s}^{-1}$$

$$\beta_n(v) = 0.125 \exp(-(v_{mV} + 65.0)/80.0) \times 10^3 \text{ s}^{-1}$$

where $v_{mV} = v_{m,2} \times 10^3$. Under direct stimulation, the isolated classical C-fiber conducts at **0.41 m/s**.

In Phase 2, the generic Na^+ current is replaced with literature-derived TTX-R Nav1.8 and Nav1.9 nociceptor kinetics (Section 2.5).

2.3 Extracellular Coupling, Regularization, and Exact Algebraic Elimination

The longitudinal extracellular cleft resistance per unit length $r_e = \rho_e / A_{\text{eff}} (\Omega/\text{m})$ is determined by extracellular resistivity ($\rho_e = 100 \Omega \cdot \text{cm} = 1.0 \Omega \cdot \text{m}$) and the annular cleft area A_{eff} :

$$A_{\text{eff}} = \pi \left[\left(\frac{d_1 + d_2}{2} + w_{\text{cleft}} \right)^2 - \left(\frac{d_1}{2} \right)^2 - \left(\frac{d_2}{2} \right)^2 \right]$$

In the zero-cleft limit ($w_{\text{cleft}} \rightarrow 0$), A_{eff} approaches $A_{\text{eff,min}} = \pi d_1 d_2 / 2 \approx 15.7 \mu\text{m}^2 = 1.57 \times 10^{-11} \text{ m}^2$, bounding r_e at an upper ceiling $r_{e,max} = \rho_e / A_{\text{eff,min}} \approx 6.37 \times 10^{10} \Omega/\text{m}$. For the nominal restricted cleft width $w_{\text{cleft}} = 20 \text{ nm}$, $A_{\text{eff}} \approx 16.4 \mu\text{m}^2 (1.64 \times 10^{-11} \text{ m}^2)$ and $r_e \approx 6.10 \times 10^{10} \Omega/\text{m}$.

To avoid numerical stiffness associated with the explicit evaluation of membrane capacitive displacement currents, we implement an algebraic elimination that incorporates the capacitive, ionic, and applied-current terms through the membrane equations while removing their explicit appearance from the extracellular solve. From core-conductor theory, total transmembrane current per unit length exiting axon k equals the axial current divergence:

$$i_{m,k}(z, t) = \frac{1}{r_{i,k}} \frac{\partial^2 u_{i,k}}{\partial z^2} = \frac{1}{r_{i,k}} \left(\frac{\partial^2 v_{m,k}}{\partial z^2} + \frac{\partial^2 u_e}{\partial z^2} \right)$$

where $r_{i,k} = 4 \rho_i / (\pi d_{axon,k}^2)$ is the intracellular longitudinal core resistance per unit length (Ω/m). Extracellular current conservation per unit length requires:

$$\frac{1}{r_e} \frac{\partial^2 u_e}{\partial z^2} - G_e u_e + \sum_k i_{m,k} = 0$$

where $G_e = \kappa / r_e$ (S/m) represents weak ohmic leakage from the restricted cleft to the surrounding bulk interstitial tissue, and κ (m^2) is a phenomenological regularization parameter. Substituting $i_{m,k}$ yields:

$$\left(\frac{1}{r_e} + \sum_k \frac{1}{r_{i,k}} \right) \frac{\partial^2 u_e}{\partial z^2} - \frac{\kappa}{r_e} u_e = - \sum_k \frac{1}{r_{i,k}} \frac{\partial^2 v_{m,k}}{\partial z^2}$$

Dividing by the total core conductance $\Sigma_G = (1)/r_e + \sum_k (1)/r_{i,k}$ produces a regularized Poisson/Helmholtz equation:

$$\frac{\partial^2 u_e}{\partial z^2} - \kappa_{\text{eff}} u_e = - \frac{1}{\Sigma_G} \sum_k \frac{1}{r_{i,k}} \frac{\partial^2 v_{m,k}}{\partial z^2}$$

The algebraic elimination removes the explicit capacitive-current feedback from the extracellular solve, avoiding that particular stiffness pathway; numerical stability was assessed empirically through the reported timestep/convergence tests.

where the effective regularization parameter κ_{eff} (m^2) is:

$$\kappa_{\text{eff}} = \frac{\kappa / r_e}{\Sigma_G}$$

We treat $\kappa = 1.0 \times 10^9 \text{ m}^2$ as a fixed phenomenological model parameter throughout all comparative analyses, defining a nominal pre-elimination spatial decay scale $\lambda_{\kappa} = 1/\sqrt{\kappa} \approx 31.62 \text{ } \mu\text{m}$. It is not an experimentally measured constant. Importantly, the effective spatial decay scale of the reduced Helmholtz operator is governed by $\lambda_{\text{eff}} = 1/\sqrt{\kappa_{\text{eff}}}$, where $\kappa_{\text{eff}} = (\kappa / r_e) / \Sigma_G$. Because total core conductance $\Sigma_G(n)$ expands with fiber count n , κ_{eff} decreases and λ_{eff} correspondingly lengthens: $\lambda_{\text{eff}} = 73.3 \text{ } \mu\text{m}$ for $n=1$, 209.7 μm for $n=10$, 329.1 μm for $n=25$, and 464.3 μm for $n=50$ (Table 4).

Discrete Current Conservation and Operator-Splitting Residual Analysis

To evaluate current conservation rigorously, we explicitly distinguish between two separate conservation metrics:

1. **Discrete Algebraic Operator Identity Residual (R_{alg}):** Evaluates whether the banded matrix solve numerically satisfies the discretized Helmholtz operator:

$$R_{\text{alg}}(z_j, t^{n+1}) = \left| \left(\frac{1}{r_e} + \sum_k \frac{1}{r_{i,k}} \right) \nabla^2 u_e^{n+1}(z_j) - \frac{\kappa}{r_e} u_e^{n+1}(z_j) + \sum_k \frac{1}{r_{i,k}} \nabla^2 v_k^n(z_j) \right|$$

Across all spatial compartments and timesteps, the discrete operator identity is satisfied to machine precision: $\max |R_{\text{alg}}| \approx 1.95 \times 10^{-17} \text{ A/m}$ ($1.3 \times 10^{-12}\%$ of peak transmembrane source current, $\approx 1.49 \times 10^{-3} \text{ A/m}$). Both residuals are shown in Fig. 3.

2. **Physical Dynamic Current-Balance Residual (R_{phys}):** Evaluates the physical Kirchhoff current balance using the actual updated transmembrane currents $i_{m,k}^{\text{phys}}(z_j, t^{n+1}) = \pi d_{\text{axon},k} [C_{m,k}(v_k^{n+1} - v_k^n)/(\Delta t) + I_{\text{ion},k}(v_k^n, w_k^{n+1}) - I_{\text{stim},k}^{n+1}]$:

$$R_{\text{phys}}(z_j, t^{n+1}) = \left| \frac{1}{r_e} \nabla^2 u_e^{n+1}(z_j) - \frac{\kappa}{r_e} u_e^{n+1}(z_j) + \sum_k i_{m,k}^{\text{phys}}(z_j, t^{n+1}) \right|$$

Because linear axial diffusion is stepped implicitly while u_e^{n+1} is computed from the explicit state v_k^n , the physical current balance exhibits an implicit–explicit operator-splitting lag:

$$L_{\text{split}}(z_j, t^{n+1}) = \sum_k \frac{1}{r_{i,k}} \nabla^2 (v_k^{n+1} - v_k^n) = O(\Delta t)$$

Under systematic temporal refinement across $\Delta t \in [5.0, 2.5, 1.25, 0.625, 0.3125, 0.15625] \mu\text{s}$ at fixed $\Delta z = 10 \mu\text{m}$, and the benchmark-default cleft width of 100 nm, the peak physical residual converges monotonically toward zero:

- $\Delta t = 5.0 \mu\text{s}$: $\max |R_{\text{phys}}| = 3.88 \times 10^{-4} \text{ A/m}$ (26.20% of peak source)
- $\Delta t = 2.5 \mu\text{s}$ (nominal production step): $\max |R_{\text{phys}}| = 3.24 \times 10^{-4} \text{ A/m}$ (21.78%)
- $\Delta t = 1.25 \mu\text{s}$: $\max |R_{\text{phys}}| = 2.59 \times 10^{-4} \text{ A/m}$ (17.35%)
- $\Delta t = 0.625 \mu\text{s}$: $\max |R_{\text{phys}}| = 1.82 \times 10^{-4} \text{ A/m}$ (12.19%)
- $\Delta t = 0.3125 \mu\text{s}$: $\max |R_{\text{phys}}| = 1.15 \times 10^{-4} \text{ A/m}$ (7.70%)
- $\Delta t = 0.15625 \mu\text{s}$: $\max |R_{\text{phys}}| = 6.65 \times 10^{-5} \text{ A/m}$ (4.44%)

The apparent local convergence order rises from $p = 0.26$ at coarse steps to $p = 0.79$ at the finest step pair (`results/convergence/kcl_refinement_study.csv`), and is still increasing at the smallest step; the data are consistent with, but do not by themselves demonstrate, first-order asymptotic convergence of the IMEX splitting lag.

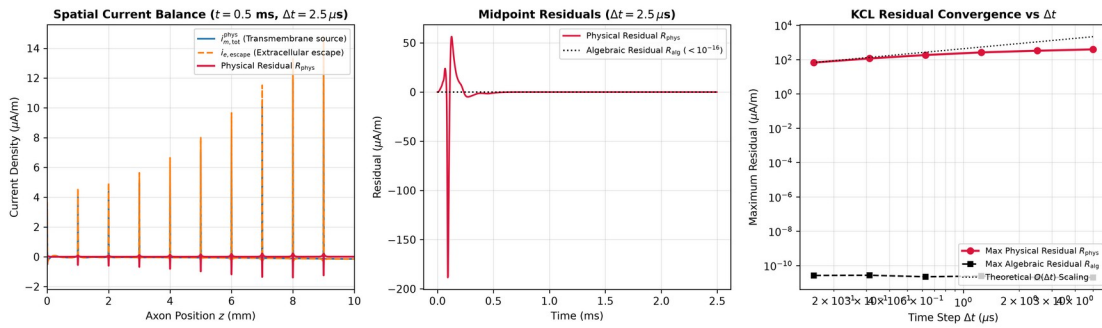


Fig. 3 Discrete Kirchhoff's current law conservation and temporal refinement. (a) Spatial profiles of dynamic transmembrane source currents and extracellular return current at peak action-potential

propagation ($t = 0.25$ ms). (b) Discrete algebraic operator identity residual R_{alg} , verified to machine precision ($< 10^{-16}$ A/m). (c) Physical current-balance residual R_{phys} convergence under temporal refinement ($\Delta t \in [5.0, 0.156]$ μ s), demonstrating monotonic reduction of the IMEX operator-splitting lag from 26.2% to 4.44% (apparent order rising from 0.26 to 0.79). Benchmark cleft width: 100 nm.

2.4 Numerical Scheme and Convergence Methodology

The system is integrated using an implicit–explicit (IMEX) operator-splitting scheme on a spatial grid of nominal $\Delta z = 10$ μ m ($N = 1000$ compartments) with time step $\Delta t = 2.5$ μ s. Linear axial diffusion terms are stepped implicitly via backward Euler using tridiagonal LAPACK solvers (`solve_banded`), while nonlinear ionic currents are updated explicitly. Extracellular potential u_e is obtained at each step by solving the banded Helmholtz operator.

2.4.1 Spatial Refinement Study

To verify spatial convergence, Δz was refined across $[20, 10, 5, 2.5, 1.25, 1.0]$ μ m holding all physical dimensions, including $l_{node} = 1.0$ μ m, strictly fixed, with the temporal step held constant at $\Delta t = 2.5$ μ s and the cleft width at $w_{cleft} = 20$ nm (`src/run_convergence_battery.py`). As detailed in Section 3.1 and Table 5, the isolated A β conduction velocity (**41.03 m/s**) and the action-potential excursion at $z = 4$ mm (**80.44 mV**, an absolute peak of +0.44 mV from rest -80.0 mV) were identical at every grid step. The coupled responses converged monotonically: the single-pair ΔV ($n=1$) rose from 1.237 mV at 20 μ m to 1.354 mV at 1.0 μ m, and the multi-fiber ΔV ($n=25$) from 5.919 mV to 6.050 mV, with successive changes below 0.5% for $\Delta z \leq 2.5$ μ m. The nominal grid ($\Delta z = 10$ μ m) underestimates the finest-grid values by 2.1% ($n=1$) and 0.6% ($n=25$).

2.4.2 Temporal Refinement Study

Temporal convergence was evaluated independently across $\Delta t \in [5.0, 2.5, 1.25, 0.625, 0.5, 0.25, 0.125]$ μ s (holding $\Delta z = 10.0$ μ m and $w_{cleft} = 20$ nm fixed). Single-pair and multi-fiber event classifications (*Local Stimulus Transient*) remain unchanged across all time steps. A β conduction velocity depends weakly on the time step: it decreases from 42.11 m/s at 5.0 μ s to 41.03 m/s at the nominal 2.5 μ s and 40.00 m/s at 1.25 μ s, then changes by less than 0.3% per halving (39.80 m/s at 0.125 μ s). However, as documented in Section 4.6, the absolute peak amplitude of the single-pulse multi-fiber ($n=25$) response exhibits notable timestep dependence: 6.015 mV at $\Delta t = 2.5$ μ s, 8.673 mV at 1.25 μ s, 11.557 mV at 0.625 μ s, 12.424 mV at 0.5 μ s, 14.470 mV at 0.25 μ s, and 15.313 mV at 0.125 μ s (`results/convergence/temporal_convergence_final.csv`). Richardson extrapolation across the finest dyadic steps ($\Delta t \in [0.5, 0.25, 0.125]$ μ s, apparent order $p \approx 1.28$) yields an asymptotic estimate of ≈ 15.90 mV (or ≈ 16.16 mV under first-order extrapolation, $p = 1.0$). We explicitly emphasize that these extrapolated values are mathematical estimates rather than directly simulated converged points; thus, while event classification remains robust and invariant across all tested timesteps, single-pulse potential magnitude is not quantitatively converged.

2.4.3 Independent Reference Solver Validation for A β Fiber

To verify the numerical implementation of the saltatory A β axon model against an independent integration scheme, a standalone reference solver was implemented (`audit_reference_solver.py`). It integrates the axial diffusion term with Crank-Nicolson and

updates the CRRSS gates with a Rush-Larsen step at the midpoint voltage, treating ionic and stimulus currents explicitly as in the production solver. Evaluated at $\Delta t = 0.625 \mu\text{s}$ and nominal $\Delta z = 10 \mu\text{m}$, with a 2-nA, 0.2-ms stimulus at node 0, the production IMEX solver yielded an A β conduction velocity of 40.00 m/s compared to 39.51 m/s for the Crank-Nicolson reference solver (a 1.2% difference; the action potential excursion at $z = 4 \text{ mm}$ differed by 0.24%, 79.98 mV vs 79.78 mV). This agreement supports the numerical integration of the A β cable dynamics specifically; the two solvers share the membrane model, geometry, and explicit ionic-current treatment, so the comparison does not constitute a full reference solver for the coupled multi-fiber system.

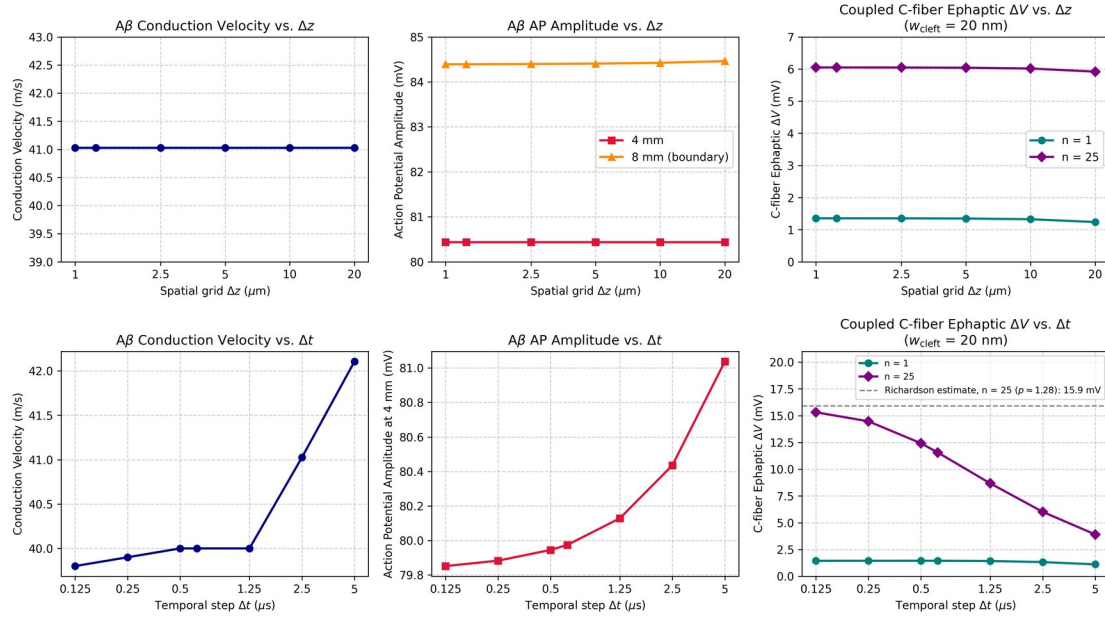


Fig. 4 Spatial and temporal convergence of the A β conduction and the coupled $n=25$ ephaptic response. Top row: spatial refinement at fixed $\Delta t = 2.5 \mu\text{s}$ (physical node width held fixed) for A β conduction velocity, A β action-potential amplitude, and coupled C-fiber ephaptic ΔV for $n = 1$ and $n = 25$ ($w_{\text{cleft}} = 20 \text{ nm}$); conduction velocity and the amplitude at 4 mm are independent of Δz , and both ΔV curves converge. Bottom row: temporal refinement at fixed $\Delta z = 10 \mu\text{m}$. Conduction velocity and amplitude decrease slightly with the time step (-5.5% and -1.5% from 5.0 to 0.125 μs) and event classification is unchanged, whereas the single-pulse $n = 25$ amplitude depends strongly on the timestep and is not converged; the dashed line marks the Richardson estimate (15.9 mV, $p \approx 1.28$).

2.5 Nav1.8/Nav1.9 Nociceptor Kinetics (Phase 2)

Phase 2 replaces the classical HH Na^+ current with TTX-R Nav1.8 and Nav1.9 currents (Akopian et al., 1996; Cummins et al., 1999; Klein et al., 2017):

$$I_{\text{ion},2} = \bar{g}_{18} m_8^3 h_8 (v_2 - E_{\text{Na}}) + \bar{g}_{19} m_9 (v_2 - E_{\text{Na}}) + \bar{g}_{\text{K}} n^4 (v_2 - E_{\text{K}}) + g_{\text{leak}} (v_2 - E_{\text{leak},2})$$

$$m_{8,\infty}(v) = \frac{1}{1 + \exp(-(v + 25.0)/6.0)}, \quad \tau_{m8} = 1.5 \text{ ms}$$

$$h_{8,\infty}(v) = \frac{1}{1 + \exp((v + 42.0)/6.0)}, \quad \tau_{h8} = 2.0 \text{ ms}$$

$$m_{9,\infty}(v) = \frac{1}{1 + \exp(-(v + 50.0)/5.0)}, \quad \tau_{m9} = 10.0 \text{ ms}$$

The delayed-rectifier K^+ gating variable n and its voltage-dependent kinetics are retained directly from the classical Hodgkin-Huxley formulation described in Section 2.2.2 (with delayed-rectifier conductance calibrated to $\bar{g}_K = 60.0 \text{ mS/cm}^2$ and $E_K = -77.0 \text{ mV}$). Nav1.8 inactivation kinetics were calibrated ($V_{1/2} = -42.0 \text{ mV}$, $k = 6.0 \text{ mV}$, $\tau_{h8} = 2.0 \text{ ms}$) to eliminate a non-repolarizing attractor and ensure physiological action potential duration (5–10 ms). Membrane parameters are $\bar{g}_{I8} = 2000 \text{ mS/cm}^2$, $\bar{g}_{I9} = 0.2 \text{ mS/cm}^2$, $E_{Na} = +50.0 \text{ mV}$, and $E_{leak,2} = -55.0 \text{ mV}$. To eliminate initialization transients, a 20-ms pre-stimulus settling simulation is performed prior to stimulus onset (`get_settled_c_fiber`), establishing a stable resting baseline at **-66.82 mV** (drift $< 10 \text{ } \mu\text{V}$ over 20 ms; steady-state gating variables $m_8 = 0.0009$, $h_8 = 0.9843$, $m_9 = 0.0334$, $n = 0.2902$).

2.6 Multi-Fiber Bundle Formulation and Temporal Jitter

In intact peripheral nerves, unmyelinated axons are enclosed within Schwann cell cytoplasm inside Remak bundles (Murinson & Griffin, 2004), isolated from myelinated A β fibers. The present geometry models a **pathologically altered or confined lesion space** (e.g., focal compression or demyelination) where denuded axolemmas share a restricted interstitial cleft ($w_{\text{cleft}} = 20 \text{ nm}$).

To evaluate spatial summation, we extend the system to n synchronized A β fibers sharing the cleft with one C-fiber. Under synchronous stimulation, identical fibers contribute symmetrically. We model a single representative A β cable and scale its current source by n . Total core conductance scales as:

$$\Sigma_G(n) = \frac{1}{r_e} + \frac{n}{r_{i,1}} + \frac{1}{r_{i,2}}$$

To assess the impact of desynchronization, we implement an asynchronous recruitment protocol where A β activation times are distributed across a uniform jitter window of 1.5 ms ($\pm 0.75 \text{ ms}$) discretized into $K = 21$ temporal phases:

$$t_{\text{stim},k} = t_{\text{pulse}} + \delta t_k, \quad \delta t_k \in [-0.75, +0.75] \text{ ms}$$

The ensemble source term is averaged across all K phases before driving the extracellular Helmholtz solve.

2.7 Operational Detection Endpoints

To maintain strict objectivity, every simulation is classified automatically by an integrated spatiotemporal waveform analyzer (`classify_waveform` in `src/phase1_classical_hh/spike_detector.py`):

1. **No Response** (code: `NO_RESPONSE`): Peak downstream depolarization $\Delta V < 0.5 \text{ mV}$.

2. **Passive Depolarization** (code: PASSIVE_DEPOLARIZATION): Downstream $\Delta V < 45$ mV or maximum rate of rise $\max(dV/dt) < 30$ V/s.
 3. **Local Stimulus Transient** (code: LOCAL_STIMULUS_TRANSIENT): Depolarization exceeds threshold proximal to the electrode ($z < 2.0$ mm) but decays below threshold in the downstream cable window ($z \in [3.0, 8.0]$ mm).
 4. **Local Aborted Spike** (code: LOCAL_ABORTED_SPIKE): Threshold (-20 mV) crossed at $z = 4.0$ mm but fails to propagate past $z = 8.0$ mm.
 5. **Propagated Action Potential** (code: PROPAGATED_ACTION_POTENTIAL): Threshold crossed sequentially at both $z = 4.0$ mm and $z = 8.0$ mm with conduction velocity within physiological bounds ($0.2 \leq CV \leq 3.0$ m/s for C-fibers).
-

2.8 Uniform Hodgkin-Huxley Kinetics Shift (Phase 1 Sensitivity Protocol)

To explore whether classical Hodgkin-Huxley dynamics could support a quiescent yet excitable sensitized state near threshold, an artificial parameter sensitivity perturbation was applied to the Phase 1 C-fiber model. The voltage-dependent rate functions $\alpha(v)$ and $\beta(v)$ for sodium and potassium gating were subjected to a uniform hyperpolarizing shift ΔV_{shift} :

$$\tilde{\alpha}_x(v) = \alpha_x(v + \Delta V_{\text{shift}}), \quad \tilde{\beta}_x(v) = \beta_x(v + \Delta V_{\text{shift}}) \quad (x \in \{m, h, n\})$$

This protocol served strictly as a parameter sensitivity exploration rather than a physiological representation of inflammatory sensitization. It evaluated whether shifting the activation window closer to the unperturbed resting potential could enable single-pulse A β ephaptic inputs to cross threshold without precipitating runaway spontaneous pacemaking. A positive ΔV_{shift} displaces every rate function toward more negative membrane potentials. For both sensitization protocols (Sections 3.3 and 3.4) the C-fiber was started at the exact equilibrium of the unperturbed membrane (-65.13 mV), the perturbation was switched on at $t = 0$, and the membrane was followed for 1 s. Because a uniform perturbation with a quiescent A β fiber leaves the cable spatially uniform, the analysis was carried out on the space-clamped membrane equations (identical rate functions and conductances) and verified on the full coupled cable at selected points (run_sensitization_phase1.py).

3 Results

3.1 Single-Fiber Validation and Positive Controls

The isolated A β fiber conducted saltatorily across all 10 nodes, reaching an absolute peak of approximately +0.44 mV from a resting potential of -80.0 mV, corresponding to an action-potential excursion of **80.44 mV**, with an isolated conduction velocity of **41.03 m/s** (Table 5). The isolated Phase 1 C-fiber conducted continuously at **0.41 m/s** with a peak amplitude of +24.7 mV (Fig. 5b). Both models confirmed independent excitability and stable resting baselines, providing a rigorous positive control.

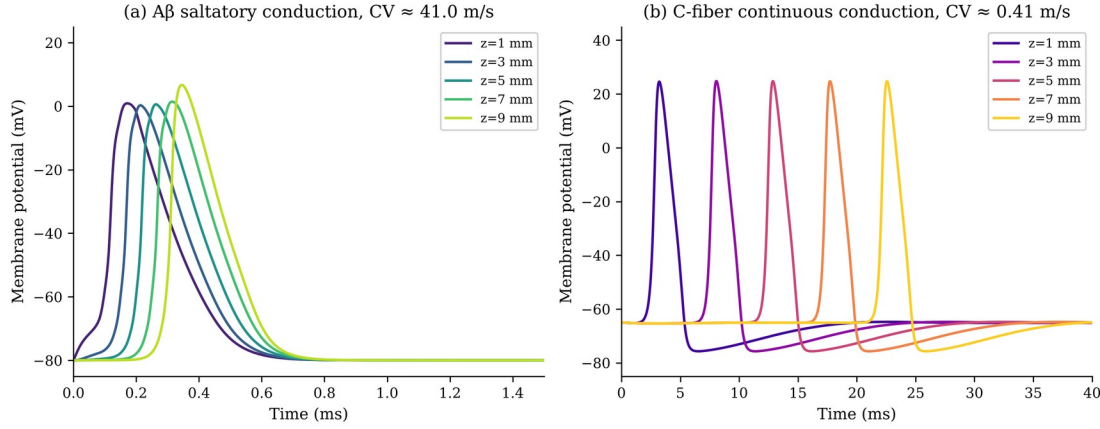


Fig. 5 Positive controls (uncoupled). (a) Isolated A β fiber: saltatory action potential (excursion 80.44 mV at $z = 4$ mm, resting at -80.0 mV and peaking at +0.44 mV) propagating at **41.03 m/s** across nodes $z = 1$ –9 mm (2 nA, 0.2 ms stimulus at $z = 0$). (b) Isolated classical HH C-fiber: regenerative action potential conducting continuously at **0.41 m/s** (1 nA, 1 ms stimulus at $z = 0$)

3.2 Experiment 1: Single-Pulse Crosstalk (Phase 1)

Across the tested cleft-width range of $w_{\text{cleft}} = 20$ nm to 5 μm , a single A β action potential failed to trigger an ectopic C-fiber action potential. At the tightest cleft (20 nm), peak C-fiber depolarization was only $\Delta V = 1.325$ mV (membrane potential rising from -65.0 mV to -63.68 mV at $z = 6.0$ mm; Table 1). The event was classified as *Local Stimulus Transient*.

3.3 Sensitization via Persistent Depolarizing Bias (Phase 1)

Applying a uniform depolarizing bias current to the C-fiber (starting from its exact resting equilibrium, -65.13 mV, with the bias switched on at $t = 0$; Section 2.8) produced a stable, quiescent sensitized state only up to a bias of ≈ 0.10 A/m 2 , where the baseline rises to -61.9 mV (0.05 A/m 2) and -59.8 mV (0.10 A/m 2). In this window the marginal ephaptic perturbation evoked by a passing A β action potential was unchanged by the bias ($\Delta V = 1.33$ mV for $n = 1$ at every bias tested) and failed to reach firing threshold. Above ≈ 0.108 A/m 2 the C-fiber fired repetitively with no A β input (60–96 Hz at 0.11–0.30 A/m 2); between 0.4 and ≈ 1.2 A/m 2 it showed large oscillations (peak-to-peak 69 mV falling to 6 mV) that stayed below 0 mV, and above ≈ 1.23 A/m 2 it settled on a stable depolarized silent state (-45.0 mV at 1.3 A/m 2 ; depolarization block) (Table 1; sensitization_phase1_bias.csv). Classical HH kinetics therefore offer only a narrow window of stable sensitization ($\lesssim 5.3$ mV of depolarization) before intrinsic pacemaking, indicating the absence of a broad stable sensitized intermediate.

3.4 Sensitization via Uniform HH Kinetics Shift (Phase 1 Sensitivity Test)

Applying a uniform shift ΔV_{shift} to all HH rate functions served as a mathematical sensitivity exploration (Section 2.8). Shifts up to 8.5 mV (and negative shifts down to -20 mV) left the fiber quiescent, with the resting potential moving slightly hyperpolarized (-66.35 mV at $\Delta V_{\text{shift}} = 3$ mV; -67.83 mV at 8.5 mV) and

the A β -evoked midpoint perturbation unchanged ($\Delta V = 1.33$ mV for $n = 1$; 5.07–5.10 mV for $n = 25$). At $\Delta V_{\text{shift}} \approx 8.56$ mV the fiber switched abruptly into spontaneous pacemaking without external stimulation (≈ 52 –58 Hz at 8.6–10 mV; `sensitization_phase1_kinetics_shift.csv`). The shift therefore did not create a stable, quiescent yet near-threshold sensitized state: the transition from quiescence to pacemaking is abrupt, which motivated Phase 2.

3.5 High-Frequency Stimulation, Propagation Timing, and Lesion Boundary Controls

Delivering pulse trains at 100 Hz ($n=25$, $w_{\text{left}}=20$ nm) reveals critical spatiotemporal propagation dynamics that distinguish stimulation-boundary artifacts from pure endogenous ephaptic crosstalk:

3.5.1 Full-Length Cable Dynamics and Propagation Latency

In the full-length coupled model ($w_{\text{left}} = 20$ nm across the entire 10-mm cable), injecting a 100-nA stimulus into 25 A β fibers at the sealed boundary ($z = 0$) generates a large local extracellular voltage transient at the boundary ($u_e \approx +265$ mV). In the Phase 1 classical Hodgkin-Huxley model, this transient initiates a slow C-fiber action potential at $t = 0.02$ ms. Because unmyelinated C-fiber conduction velocity is 0.41 mm/ms, this action potential requires:

- **12.90 ms** to propagate from $z = 0$ to the cable midpoint ($z = 5.0$ mm), and
- **20.16 ms** to reach the downstream recording electrode ($z = 8.0$ mm).

This propagation latency explains key reporting differences across historical simulation runs:

1. **Short Observation Windows ($T \leq 5.2$ ms):** In single-pulse simulations evaluated over 5.2 ms (e.g., `EXP_E02_E03_reconciliation.csv` and `phase2_temporal_summation.csv`), the traveling wave has only propagated to $z \approx 1.8$ mm. Downstream recording sites ($z \geq 3.0$ mm) record only the rapid passing A β wave at $t \approx 0.25$ ms (+5.039 mV at midpoint, +6.015 mV downstream maximum at $z = 7.0$ mm). Consequently, the spike classifier correctly classifies the downstream trace as *Local Stimulus Transient* with `c_propagated_ap = False`.
2. **Extended Observation Windows ($T \geq 30$ ms):** When the single-pulse simulation is extended to $T = 30$ ms (`final_100Hz_reconciliation.csv`, row 0), the slow wave initiated at $t=0$ completes its traverse, reaching the downstream electrode at $t = 20.16$ ms ($V_2 = +24.74$ mV, `spiked = True`).
3. **100-Hz Pulse Train Progression:** In a 100-Hz train (inter-pulse interval 10 ms), Pulse 2 is delivered at $t = 10$ ms. The traveling spike initiated at $t=0$ reaches the midpoint at $t = 12.90$ ms—within the Pulse 2 observation window ([10, 20] ms)—producing a midpoint peak of +24.83 mV. In historical exploratory runs truncated at $T = 15.2$ ms, the wave was captured in flight at $z \approx 2.1$ mm before reaching the downstream electrode ($z = 8.0$ mm), producing an artificial *Local Aborted Spike* classification. With adequate observation time ($T \geq 35$ ms), fully propagated action potentials are recorded downstream (midpoint peak +24.83 mV, downstream peak +24.70 mV at 8.0 mm, $CV = 0.41$ m/s; *Propagated Action Potential*, `spiked = True`; Table 1).
4. **State-Reset Mechanism Control:** Resetting the C-fiber membrane potential and gating variables to rest immediately prior to Pulse 2 ($t = 10$ ms; `run_test3_mechanism.py`) physically extinguishes the traveling wave in flight at $z \approx 4.0$ mm, completely preventing downstream spike

arrival. This confirms that downstream threshold crossing in repetitive trains requires uninterrupted continuous membrane-state propagation along the unmyelinated cable.

3.5.2 Grounded Stimulation Site and Interior Lesion Control

To determine whether ephaptic threshold crossing can occur purely from endogenous A β action potentials traveling through a pathological lesion without any boundary stimulus coupling, an interior lesion control experiment was implemented (`src/interior_lesion_test.py` and `src/run_lesion_audit.py`). In this control:

- The stimulation site ($z \in [0, 2.0]$ mm) is grounded in bulk interstitial fluid ($r_e = 0$, $u_e = 0$).
- The restricted pathological cleft ($w_{\text{cleft}} = 20\text{--}50$ nm) is strictly confined to an interior lesion zone ($z \in [3.0, 7.0]$ mm).
- A β action potentials are initiated in the grounded zone and propagate endogenously into the lesion.

Under these conditions, pure endogenous ephaptic crosstalk produced peak C-fiber depolarizations of only 1.33 mV in Phase 1 (single A β fiber, 20-nm cleft; peak -63.67 mV against a -65.0 mV rest) and 9.48 mV in Phase 2 ($n=25$, 20-nm cleft; peak -57.41 mV) inside the lesion. **No action potential was evoked in the C-fiber in either phase.** In Phase 1 the lesion peak was identical for 5, 10 and 20 pulses at 800 Hz, and in Phase 2 a 10-pulse 800-Hz train gave the same -57.41 mV peak as a single pulse, so no pulse-to-pulse build-up was observed (`run_lesion_audit.py`); 20-nm and 50-nm clefts differed by about 0.1 mV or less. Peak depolarization therefore remained strictly subthreshold in this grounded configuration. The control covers one geometry (grounded stimulation site, lesion at 3–7 mm, 20–50 nm clefts, 800-Hz trains) and does not exclude threshold crossing under other conditions.

3.5.3 Threshold Resistance of Nav1.8/Nav1.9 Nociceptors (Phase 2)

Furthermore, in the Phase 2 nociceptor model incorporating realistic TTX-resistant Nav1.8 and Nav1.9 kinetics, the higher activation threshold of Nav1.8 (-35.0 mV vs -55.0 mV for classical HH Na⁺) renders the C-fiber resistant to threshold crossing at moderate recruitment ($n \leq 28$; at $n \geq 29$ the ungrounded boundary transient can launch a spike, Section 3.9.1):

- Even in the ungrounded full-length cleft with $n = 25$ synchronized A β fibers and extended 40-ms simulation time, the Phase 2 C-fiber reaches a midpoint peak of only **-56.66 mV** ($\Delta V = 10.16$ mV above resting baseline -66.82 mV; `phase2_navc_multifiber.csv`).
- The peak remains 21.66 mV below the Nav1.8 threshold (-35.0 mV), and in the single-pulse and $n \leq 28$ conditions of Fig. 7 the C-fiber produces `c_propagated_ap = False` and `spiked = False`. A 40-ms run at $n = 28$ gives the same result (midpoint peak -56.51 mV; no C-fiber voltage above -58.8 mV at $z \geq 0.3$ mm after 1.5 ms).

3.6 Regularization Sensitivity Analysis (κ)

A lower- κ sensitivity exploration ($\kappa = 3 \times 10^7 \text{ m}^{-2}$, $\lambda_e \approx 183 \text{ }\mu\text{m}$) produced nonphysical delayed threshold crossings under the tested numerical implementation. Because this behavior was diagnosed as numerical instability under the tested lower- κ implementation rather than an independently motivated physiological parameter choice or physical stability boundary, it is not used to support biological conclusions; all

comparative analyses across the paper therefore retain the prespecified phenomenological baseline $\kappa = 1.0 \times 10^9 \text{ m}^{-2}$.

3.7 Nav1.8/Nav1.9 C-Fiber Positive Controls and Baseline (Phase 2)

Following 20 ms settling, the uncoupled Nav1.8/Nav1.9 C-fiber rested stably at **-66.82 mV** with baseline drift $< 10 \mu\text{V}$ (steady-state gating variables $m_8 = 0.0009$, $h_8 = 0.9843$, $m_9 = 0.0334$, $n = 0.2902$). Direct current injection (2 nA, 1 ms) evoked full-amplitude continuous action potentials propagating at **0.26 m/s** (Fig. 6a), providing a positive-control demonstration of excitability and propagation within the Phase 2 model.

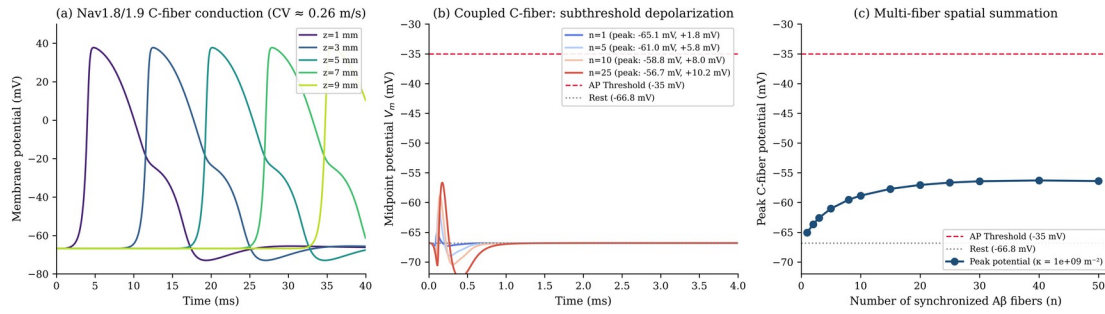


Fig. 6 Phase 2 positive control and multi-fiber summation. (a) Isolated Nav1.8/Nav1.9 C-fiber action potential conducting continuously at 0.26 m/s ($z = 1\text{--}9 \text{ mm}$). (b) Subthreshold C-fiber midpoint membrane potential traces for $n \in [1, 5, 10, 25]$ synchronized Aβ fibers ($w_{\text{cleft}} = 20 \text{ nm}$). (c) Multi-fiber spatial summation trend: peak C-fiber midpoint potential as a function of synchronized Aβ fiber count $n \in [1, 50]$, rising from -65.1 mV at $n = 1$ to -56.7 mV at $n = 25$, showing an overall increase with a small fluctuation at the largest n (-56.29 mV at $n = 40$; -56.40 mV at $n = 50$) and saturating near -56.4 mV due to core conductance shunting (Σ_C)

3.8 Phase 2 Single-Pair Crosstalk

Single-pulse stimulation across $w_{\text{cleft}} \in [20 \text{ nm}, 5 \mu\text{m}]$ in the Nav1.8/Nav1.9 model confirmed the Phase 1 negative finding: no action potentials were evoked, with peak depolarization of 1.76 mV at 20 nm, falling monotonically to 0.12 mV at 5 μm . Graded depolarizing bias currents (0–2 A/m²) shifted the C-fiber baseline smoothly from -66.82 mV to -59.30 mV at 0.25 A/m² and -47.39 mV at 2 A/m², with the membrane remaining on a stable quiescent equilibrium at every bias up to $\approx 3.2 \text{ A/m}^2$ (Nav1.8 availability h_8 falling from 0.98 to 0.71 over 0–2 A/m²). The Aβ-evoked perturbation barely changed ($\Delta V = 1.71\text{--}1.76 \text{ mV}$ for $n = 1$ and 9.5–10.2 mV for $n = 25$ at the midpoint) and remained well below the Nav1.8 activation threshold (Table 2; run_sensitization_phase2.py). Depolarization block, a jump to a silent, Nav1.8-inactivated state near -21 mV ($h_8 \approx 0.03$), occurred only above $\approx 3.2 \text{ A/m}^2$.

3.9 Multi-Fiber Bundle Spatial Summation, Temporal Jitter, and Mechanism Control

3.9.1 Multi-Fiber Spatial Summation

In the Phase 1 classical HH model, sweeping synchronized A β fiber count $n \in [1, 50]$ in a 20 nm cleft ($V_{rest} = -65.000$ mV) produced a progressive increase in subthreshold depolarization: from $\Delta V = 1.325$ mV at $n = 1$ to $\Delta V = 5.039$ mV at midpoint (and 6.015 mV downstream maximum) at $n = 25$ (Table 4). As n increased further across the tested range up to 50, depolarization exhibited saturation-like behavior, leveling off at $\Delta V = 6.184$ mV. This plateau is directly attributable to the expanding parallel intracellular core conductance $\Sigma_G(n)$, which shunts the extracellular cleft. Concurrently, collective cleft loading decelerates A β conduction velocity from 40.0 m/s ($n = 1$) to **32.00 m/s** ($n = 25$) and 27.1 m/s ($n = 50$). Across the entire tested range of $n \leq 50$, single-pulse multi-fiber spatial summation remains strictly subthreshold.

In the Phase 2 Nav1.8/Nav1.9 nociceptor model ($V_{rest} = -66.82$ mV; Fig. 7), synchronous multi-fiber activation similarly produces an asymptotic subthreshold depolarization trend, rising from a midpoint peak of -65.07 mV ($\Delta V = 1.76$ mV) at $n = 1$ to -56.66 mV ($\Delta V = 10.16$ mV) at $n = 25$, and saturating near -56.40 mV ($\Delta V = 10.42$ mV) at $n = 50$, remaining well below the nociceptor threshold (-35.0 mV).

The midpoint values above are read within the first 3–5 ms after the pulse. In the ungrounded full-length model, the same boundary transient that produces the 100-Hz result (Section 3.5.1) can also launch a slow C-fiber action potential at the sealed boundary $z = 0$ once enough A β fibers are summed: this occurs for $n \geq 2$ in Phase 1 ($n = 1$ does not fire) and for $n \geq 29$ in Phase 2 (threshold $n^* \approx 28.5$; boundary_spike_phase1.csv, boundary_spike_phase2.csv). The spike conducts at 0.41 m/s (Phase 1) or 0.26 m/s (Phase 2) and reaches the midpoint only after roughly 13 ms (Phase 1) or 20 ms (Phase 2), so it lies outside the midpoint window and is not part of the saturation trend. In the grounded interior-lesion control no action potential occurred for any n up to 200 (Phase 2, 20 nm cleft: lesion-zone peak -56.68 mV at $n = 50$ and -58.49 mV at $n = 200$; lesion_control_high_n.csv).

3.9.2 Temporal Dispersion / Jitter Attenuation

Introducing a modeled 1.5 ms temporal onset-dispersion window across the A β population desynchronized the source currents, reducing temporal overlap and sharply attenuating depolarization. In the Phase 1 model at $n = 25$, peak downstream depolarization is suppressed from 6.015 mV to **0.426 mV**—a **92.9% attenuation**. In the Phase 2 model at $n = 25$ (Fig. 7), peak midpoint potential is suppressed from -56.66 mV ($\Delta V = 10.16$ mV) to **-66.20 mV** ($\Delta V = 0.63$ mV above rest -66.82 mV)—a **93.8% attenuation**. This establishes that multi-fiber spatial summation is highly sensitive to input synchrony under tested conditions across both channel models.

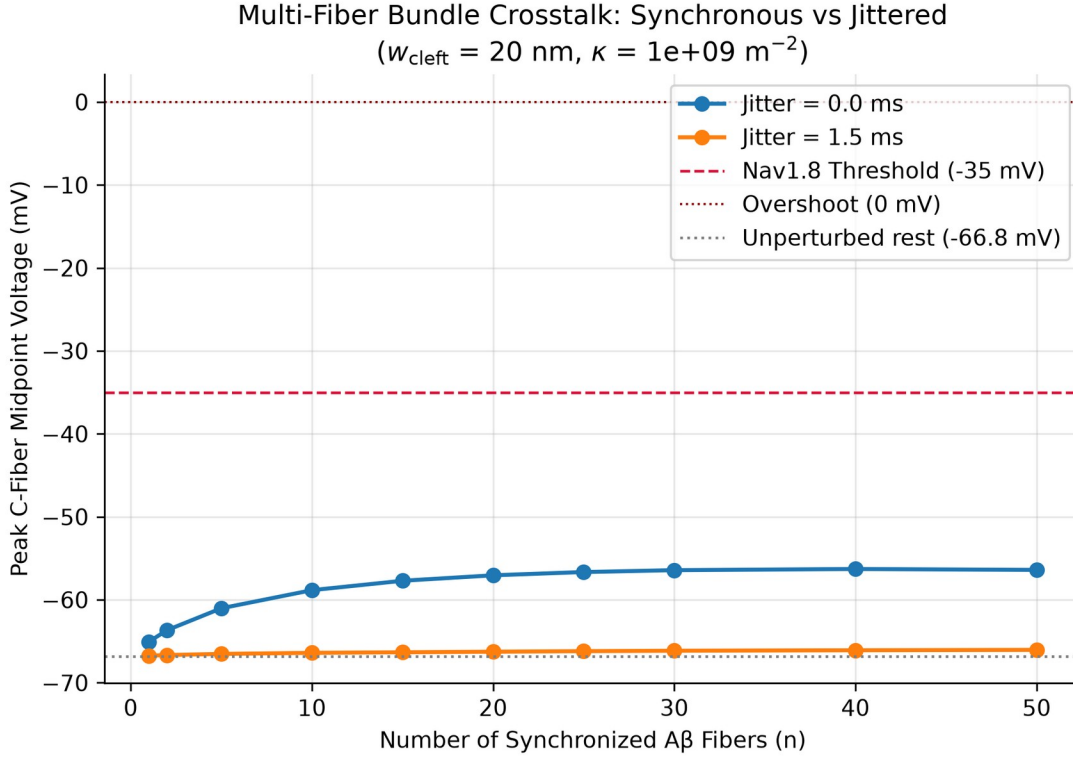


Fig. 7 Multi-fiber spatial summation and temporal dispersion in the Phase 2 Nav1.8/Nav1.9 nociceptor model. Parameters: $w_{\text{cleft}} = 20 \text{ nm}$, fixed baseline $\kappa = 1.0 \times 10^9 \text{ m}^{-2}$. Peak absolute C-fiber midpoint membrane potential (V_2 at $z = 5.0 \text{ mm}$) is plotted across synchronized Aβ fiber count $n \in [1, 50]$ under synchronous recruitment (jitter 0.0 ms; peaking at -65.07 mV for $n=1$, -58.84 mV for $n=10$, -56.66 mV for $n=25$, and -56.40 mV for $n=50$, demonstrating an asymptotic subthreshold trend) and under a modeled 1.5-ms onset-dispersion window (jitter 1.5 ms across $K=21$ phases; peaking at -66.20 mV for $n=25$, representing $\Delta V = 0.63 \text{ mV}$ above rest and a 93.8% attenuation of ephaptic depolarization). Both curves originate from the settled resting baseline of -66.82 mV (dotted gray line) and remain well below the Nav1.8 regenerative threshold (-35.0 mV, dashed crimson line)

3.9.3 Mechanism Control: Membrane State vs Extracellular Potential Accumulation

To determine whether repeated 100-Hz events are driven by extracellular potential buildup or by C-fiber membrane-state accumulation, we executed a controlled state-reset experiment (`run_test3_mechanism.py`). In the control simulation, extracellular potential pulses u_e returned to baseline between pulses, with no progressive accumulation. Resetting the C-fiber membrane potential and gating variables to rest immediately before Pulse 2 ($t = 10 \text{ ms}$) extinguished the traveling wave in flight, so no downstream spike was recorded. This shows that the downstream spike depends on uninterrupted propagation of a wave already underway; because a reset removes a wave in flight by construction, it does not by itself separate membrane-state accumulation from conduction delay. The

stronger evidence is the single-pulse run, which reproduces the same downstream spike when observed for $T \geq 30$ ms (Table 1). Together, these results indicate that the events recorded in the Pulse 2 window are delayed propagation of a spike initiated by Pulse 1 at the sealed boundary, not demonstrated pulse-to-pulse recruitment.

4 Discussion

4.1 Principal Findings

This study provides a rigorous, closed-loop computational evaluation of bidirectional ephaptic crosstalk between myelinated A β afferents and unmyelinated C-fibers. Across both classical HH and nociceptor-realistic Nav1.8/Nav1.9 channel kinetics, our simulations demonstrate:

1. **Single-pair coupling under single-pulse stimulation is insufficient under tested conditions** to evoke an action potential across the tested cleft-width range of 20 nm to 5 μ m.
2. In the Phase 1 classical-HH model, synchronized multi-fiber spatial summation produces substantial subthreshold depolarization (+5.04 to +6.01 mV for $n=25$), but plateaus asymptotically near +6.2 mV due to parallel axoplasmic core conductance shunting. The Phase 2 Nav1.8/Nav1.9 model shows the same qualitative saturation, reaching +10.16 mV at $n=25$ and +10.42 mV at $n=50$.
3. **Temporal dispersion strongly attenuates spatial summation**, reducing peak ephaptic drive by 92.9% in Phase 1 (93.8% in Phase 2) under the modeled 1.5-ms onset-dispersion window.
4. **Repetitive 100-Hz stimulation did not demonstrate temporal summation to threshold.** In the ungrounded model, the propagated spike recorded in the Pulse 2 window is initiated by Pulse 1 at the sealed stimulation boundary and delayed by slow conduction (0.41 m/s); in the grounded interior-lesion control, crosstalk remained subthreshold in every configuration tested.

4.2 Single-Pair Coupling Insufficiency

Single-pair ephaptic drive generates peak membrane perturbations of only ≈ 1.3 mV in a 20-nm cleft, substantially below the depolarization required to initiate regenerative firing under the tested Phase 1 model. Within the present core-conductor formulation, the single-fiber source produces only limited current transfer into the shared extracellular pathway. This finding holds across generic and nociceptor-specific kinetics.

4.3 Multi-Fiber Spatial Summation and Asymptotic Saturation

In the Phase 1 classical-HH model, extending the system to 25 synchronized A β fibers generates an approximately 6.0-mV depolarization (5.039 mV midpoint, 6.015 mV downstream maximum). However, the response saturates asymptotically near +6.2 mV. The Phase 2 Nav1.8/Nav1.9 model reproduces the same saturation mechanism qualitatively, reaching a midpoint depolarization of 10.16 mV at $n=25$ and 10.42 mV at $n=50$. The core-conductor derivation reveals the mathematical mechanism underlying this plateau, which involves three interacting factors: (1) as synchronized fiber count n increases, the total axial core conductance $\Sigma_G(n) = 1/r_e + n/r_{i,1} + 1/r_{i,2}$ expands linearly with each added intracellular channel;

(2) this expansion of Σ_G directly reduces the effective regularization parameter $\kappa_{\text{eff}}(n) = (\kappa/r_e)/\Sigma_G(n)$, systematically lengthening the effective electrotonic spatial decay scale of the reduced 1-D Helmholtz operator, $\lambda_{\text{eff}} = 1/\sqrt{(\kappa_{\text{eff}})}$ (from 73.3 μm at $n=1$ to 464.3 μm at $n=50$; Table 4); and (3) in the reduced extracellular operator, the expanding parallel intracellular conductances act as an intrinsic low-resistance resistive shunt that dissipates return currents longitudinally rather than confining them to the cleft, causing the induced extracellular and transmembrane potentials to plateau asymptotically. Thus, multi-fiber spatial summation alone cannot drive single-pulse suprathreshold activation under the tested geometry.

4.4 High-Frequency Stimulation, Propagation Timing, and Lesion Boundary Controls

In the ungrounded, full-length coupled model ($w_{\text{cleft}} = 20$ nm from $z = 0$ to 10 mm), the propagated C-fiber action potential recorded in the Pulse 2 window is initiated by Pulse 1 at the sealed boundary. Spatiotemporal analysis reveals that injecting 100 nA into 25 A β fibers at the sealed boundary $z = 0$ generates a large initial extracellular transient ($u_e \approx +265$ mV) that initiates a slow C-fiber action potential at $t = 0.02$ ms. Because C-fiber conduction velocity is 0.41 mm/ms, this action potential requires 12.90 ms to reach the midpoint ($z = 5.0$ mm)—falling squarely within the Pulse 2 observation window ([10, 20] ms). A single pulse observed for $T \geq 30$ ms yields the same spike, and the state-reset control shows that it depends on continuous membrane-state propagation along the unmyelinated cable, as resetting the cable at $t = 10$ ms extinguishes the traveling wave in flight at $z \approx 4.0$ mm. These results do not show that Pulse 2 or subthreshold accumulation is required for the spike.

Crucially, when the stimulation site ($z \in [0, 2.0]$ mm) is grounded in bulk fluid ($r_e = 0$, $u_e = 0$) and cleft restriction is confined to an interior lesion ($z \in [3.0, 7.0]$ mm), **no action potential is ever triggered** across Phase 1, Phase 2, or 800-Hz trains up to 20 pulses (`run_lesion_audit.py`). Furthermore, in Phase 2 with realistic TTX-resistant Nav1.8 and Nav1.9 kinetics, the high threshold of Nav1.8 (-35.0 mV) keeps the midpoint response subthreshold at $n = 25$ even in the ungrounded full-length setup (midpoint peak -56.66 mV; $\Delta V = 10.16$ mV; 21.66 mV subthreshold), and the boundary-initiated spike appears only for $n \geq 29$ (versus $n \geq 2$ in Phase 1; Section 3.9.1). Pure endogenous ephaptic crosstalk in an interior lesion is therefore strictly subthreshold across both classical and nociceptor-realistic kinetics.

4.5 Synchrony Dependence

The strong attenuation observed under the modeled 1.5 ms onset-dispersion window (92.9% in Phase 1; 93.8% in Phase 2) highlights that multi-fiber spatial summation is highly sensitive to the degree of temporal alignment across fibers. While highly synchronous activation may occur during rapid electrical stimulation or sharp mechanical transients, naturalistic tactile stimuli recruiting dispersed afferent populations would exhibit reduced temporal overlap of source currents, substantially attenuating the ephaptic potential.

4.6 Numerical and Modeling Limitations

1. **One-Dimensional Approximation:** The 1D core-conductor model longitudinally averages extracellular potential, omitting three-dimensional cleft tortuosity, micro-domains, and non-

uniform axon packing. Fully coupled 3D EMI formulations (Jæger & Tveito, 2026) are required to resolve localized field hotspots.

2. **Phenomenological Regularization (κ):** The parameter $\kappa = 1.0 \times 10^9 \text{ m}^{-2}$ represents cleft leakage but is not directly measured experimentally. Furthermore, κ_{eff} inherently depends on fiber count n through Σ_G .
 3. **Temporal Non-Convergence of Single-Pulse $n=25$ Peak:** While event classifications are strictly stable under temporal refinement down to $\Delta t = 0.125 \mu\text{s}$ (*Local Stimulus Transient* throughout), the single-pulse $n=25$ peak ΔV magnitude exhibits notable timestep dependence (6.015 mV at $\Delta t = 2.5 \mu\text{s}$ to 15.313 mV at $0.125 \mu\text{s}$, $w_{\text{cleft}} = 20 \text{ nm}$). Richardson extrapolation across the finest steps yields an asymptotic estimate of $\approx 15.90 \text{ mV}$ (first-order $\approx 16.16 \text{ mV}$), but these are extrapolated estimates rather than directly simulated converged points. Absolute potential magnitudes must therefore be interpreted with caution, whereas threshold-crossing and propagation classifications remain robust.
 4. **Boundary Stimulation vs. Interior Lesion Coupling:** In the full-length coupled model, current injection at the sealed boundary $z = 0$ creates a localized extracellular potential transient that can initiate a boundary spike (Phase 1: $n \geq 2$; Phase 2: $n \geq 29$; Section 3.9.1). When the stimulation site is grounded and cleft restriction is confined to an interior lesion ($z \in [3, 7] \text{ mm}$), pure endogenous ephaptic crosstalk remained subthreshold (peak depolarization 1.33 mV in Phase 1 and 9.48 mV in Phase 2; no action potentials, including under 800-Hz trains). This control covers one geometry and a 20–50 nm cleft range, so it does not establish that threshold crossing is impossible in other configurations.
 5. **Reduced Multi-Fiber Scaling:** The multi-fiber model uses a 1D synchronized source-scaling approximation rather than an explicit 3D anatomical bundle reconstruction.
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4.7 Biological and Clinical Implications

These computational findings generate testable biophysical hypotheses, reframing peripheral ephaptic interaction as a **subthreshold depolarizing influence** rather than an autonomous single-pulse spark generator under the modeled conditions. In pathological scenarios where nociceptors may already be depolarized by inflammatory mediators, subthreshold ephaptic depolarization from synchronized A β input could plausibly bring a nociceptor closer to threshold; the present simulations do not show threshold crossing by this route, and the direct etiology of clinical dynamic mechanical allodynia in vivo remains to be established experimentally.

4.8 Future Work

Key future directions include: (1) extending the model to higher-dimensional parallel axon networks (Chawla & Morgera, 2024; Reutskiy et al., 2003) and 3D EMI finite-element models (Jæger & Tveito, 2026); (2) driving the bundle with naturalistic human microneurography spike trains; and (3) incorporating detailed kinetic models of Nav1.7 and subthreshold potassium channels (Pelot et al., 2021; Sundt et al., 2015; Tigerholm et al., 2014).

5 Conclusion

A closed-loop core-conductor model demonstrates that single-pair ephaptic crosstalk under single-pulse stimulation is insufficient to trigger ectopic nociceptor action potentials across both classical HH and nociceptor-realistic Nav1.8/Nav1.9 kinetics. Synchronized multi-fiber spatial summation produces substantial subthreshold depolarization (in Phase 1: +5.04 mV midpoint, +6.01 mV downstream maximum for $n=25$ at the nominal production timestep; in Phase 2: +10.16 mV midpoint depolarization to -56.66 mV) that asymptotically saturates due to core conductance shunting, and is strongly attenuated by temporal dispersion (92.9% attenuation in Phase 1, 93.8% in Phase 2). In the ungrounded full-length model, a 100-Hz train yields a propagated C-fiber action potential during the second pulse interval, but this spike is initiated by the first pulse at the sealed stimulation boundary and delayed by slow conduction (≈ 0.41 m/s); a single pulse reproduces it, and the state-reset control shows that it requires uninterrupted propagation. In a grounded interior-lesion control, crosstalk remained subthreshold in every configuration tested. Within the parameter space explored, ephaptic coupling in restricted nerve clefts therefore produced subthreshold depolarization but did not by itself cross threshold; whether other conditions could do so remains untested.

Nomenclature

Table 3 Definition of symbols, parameters, and SI units used in the core-conductor formulation.

Symbol	Definition	Physical Units
$V_{m,k}$	Transmembrane potential of fiber k ($k=1$: A β , $k=2$: C-fiber)	V or mV
$U_{i,k}$	Intracellular potential of fiber k	V or mV
U_e	Shared extracellular cleft potential	V or mV
$r_{i,k}$	Longitudinal intracellular core resistance per unit length	Ω/m
r_e	Longitudinal extracellular cleft resistance per unit length	Ω/m
Σ_G	Total longitudinal core conductance per unit length ($1/r_e + \Sigma 1/r_{i,k}$)	$S \cdot m$ or m/Ω
κ	Phenomenological extracellular leakage regularization parameter	m^{-2}
κ_{eff}	Effective regularization parameter ($(\kappa/r_e)/\Sigma_G$)	m^{-2}
w_{cleft}	Extracellular cleft width	nm or m
D_1	Outer fiber diameter of myelinated A β axon (10.0 μm)	m
$d_{axon,k}$	Inner axonal core diameter ($d_{axon,1}=7.0 \mu m$, $d_{axon,2}=1.0 \mu m$)	m
l_{node}	Fixed physical length of node of Ranvier (1.0 μm)	m
ρ_i, ρ_e	Axoplasmic (0.547 $\Omega \cdot m$) and extracellular (1.0 $\Omega \cdot m$) bulk resistivity	$\Omega \cdot m$
$C_{m,k}$	Specific membrane capacitance per unit inner area	F/m^2
$I_{ion,k}, I_{stim,k}$	Transmembrane ionic and injected stimulation current density	A/m^2
$\bar{g}_{Na}, \bar{g}_{K}, g_{leak}$	Maximum ionic conductance densities (CRRSS / classical HH)	S/m^2 or mS/cm^2
$\bar{g}_{18}, \bar{g}_{19}$	Maximum Nav1.8 / Nav1.9 conductance densities	S/m^2 or mS/cm^2
n	Number of synchronously active A β fibers in multi-fiber bundle	dimensionless
$\Delta z, \Delta t$	Spatial and temporal discretization steps	m, s ($\mu m, \mu s$)

Experimental Summary Tables

Table 1 Phase 1 (classical HH C-fiber): summary of experimental conditions and outcomes.

Condition	Maximum ΔV (mV)	Propagated AP?	Event Classification	Notes
Single pulse ($w_{\text{left}} = 20 \text{ nm}$ to $5 \mu\text{m}$)	1.325	False	<i>Local Stimulus Transient</i>	Subthreshold perturbation; decays passively downstream.
Mild tonic bias ($\leq 0.10 \text{ A/m}^2$) + pulse	1.33	False	<i>Local Stimulus Transient</i>	Stable quiescent baseline (-61.9 mV at 0.05 , -59.8 mV at 0.10 A/m^2); evoked response independent of bias ($n = 1$).
Aggressive tonic bias ($\geq 0.11 \text{ A/m}^2$)	N/A	Spontaneous	<i>Intrinsic Pacemaking</i>	Repetitive firing from $\approx 0.108 \text{ A/m}^2$ ($60\text{--}96 \text{ Hz}$ at $0.11\text{--}0.30$); large sub- 0 mV oscillations at $0.4\text{--}1.2$; stable depolarization block ($\approx -45 \text{ mV}$) above $\approx 1.23 \text{ A/m}^2$.
Gating kinetics shift ($\Delta V_{\text{shift}} \geq 8.6 \text{ mV}$)	N/A	Spontaneous	<i>Intrinsic Pacemaking</i>	Quiescent for all smaller shifts (rest moves to -66 to -68 mV); abrupt onset of pacemaking at $\approx 8.56 \text{ mV}$ (sensitivity test).
100 Hz train (1 pulse, $n=25$)	5.039 (mid), 6.015 (max)	False ($T \leq 5.2 \text{ ms}$) / True ($T \geq 30 \text{ ms}$)	<i>Local Stimulus Transient</i> ($T \leq 5.2 \text{ ms}$)	In short windows ($T \leq 5.2 \text{ ms}$), downstream electrodes record only subthreshold passing $A\beta$ wave. In extended windows ($T \geq 30 \text{ ms}$), sealed-boundary spike initiated at $z=0$ arrives downstream at $t = 20.16 \text{ ms}$ ($+24.74 \text{ mV}$).
100 Hz train (2 pulses, $n=25$)	$+24.83$ (mid), $+24.70$ (8 mm)	True	<i>Propagated Action Potential</i>	AP initiated by Pulse 1 arrives during the Pulse 2 window (conduction delay; a single pulse yields the same spike); requires $T \geq 35 \text{ ms}$.
100 Hz train (3, 5, 10 pulses, $n=25$)	$+24.83$ (mid), $+24.70$ (8 mm)	True	<i>Propagated Action Potential</i>	Successive spikes consistent with boundary initiation by each pulse plus

Condition	Maximum ΔV (mV)	Propagated AP?	Event Classification	Notes
Interior lesion control (grounded $z \in [0, 2]$ mm, lesion $z \in [3, 7]$ mm; Phase 1: single Aβ fiber, 5–20 pulses at 800 Hz; Phase 2: $n=25$, 10 pulses)	1.33 (Phase 1); 9.48 (Phase 2)	False	<i>Local Stimulus Transient</i>	conduction delay ($CV=0.41$ m/s); pulse-to-pulse dependence not separately tested. Stimulation site grounded in bulk fluid. Pure endogenous ephaptic crosstalk in the focal lesion remained subthreshold in every case tested; no spike was triggered, and the lesion peak did not grow with pulse count.
Lower-κ sensitivity check ($\kappa = 3 \times 10^7$ m⁻²)	N/A	False	<i>Numerical Artifact</i>	Numerical instability under lower- κ regularization; delayed and geometry-independent.

Table 2 Phase 2 (Nav1.8/Nav1.9 C-fiber): summary of experimental conditions and outcomes.

Condition	Maximum ΔV (mV)	Propagated AP?	Event Classification	Notes
Single pulse ($w_{\text{cleft}} = 20$ nm to 5 μm)	1.76	False	<i>Local Stimulus Transient</i>	Strictly subthreshold across the tested cleft-width range (20 nm to 5 μm).
Graded bias (≤ 2 A/m²)	Stable baseline	False	<i>Quiescent Sensitized</i>	Stable sensitized resting window (-59.30 mV at 0.25 A/m ² ; -47.39 mV at 2 A/m ²).
Bias + pulse (0.25–2 A/m²)	1.71–1.75	False	<i>Local Stimulus Transient</i>	Marginal ephaptic perturbation remains subthreshold; barely changed by bias ($n = 1$).
Depolarizing block ($\gtrsim 3.2$ A/m²)	N/A	False	<i>Depolarizing Block</i>	Sustained silent state (≈ -21 mV; Nav1.8 availability $h_\infty \approx 0.03$) reached from rest only above ≈ 3.2

Condition	Maximum ΔV (mV)	Propagated AP?	Event Classification	Notes
Multi-fiber bundle ($n = 25$, synchronous)	10.16 (mid peak -56.66 mV)	False	<i>Local Stimulus Transient</i>	A/m ² . Subthreshold spatial summation; plateaus near -56.4 mV at $n = 50$ ($\Delta V \approx 10.4$ mV; rest -66.82 mV). A boundary-initiated spike for $n \geq 29$ arrives after the 5-ms window (Sec. 3.9.1).
Multi-fiber bundle ($n = 25$, 1.5 ms jitter)	0.63 (mid peak -66.20 mV)	False	<i>Local Stimulus Transient</i>	93.8% attenuation due to reduced temporal overlap of source currents (rest - 66.82 mV).

Table 4 Multi-fiber spatial summation sweep in the Phase 1 classical Hodgkin-Huxley model across synchronized A β fiber count $n \in [1, 50]$ ($w_{\text{left}} = 20$ nm, fixed baseline $\kappa = 1.0 \times 10^9$ m⁻², $V_{\text{rest}} = -65.000$ mV). Values are read within the first 5 ms; in the ungrounded model a boundary-initiated C-fiber spike appears later for $n \geq 2$ (Section 3.9.1).

Fiber Count (n)	Downstream Peak ΔV (mV)	Downstream Peak V_2 (mV)	Midpoint Peak V_2 (mV)	A β CV (m/s)	Effective κ_{eff} (m ⁻²)	Effective Length Scale λ_{eff} (μm)	Classification
1	1.325	-63.675	-63.712	40.00	1.86×10^8	73.3	<i>Local Stimulus Transient</i>
2	2.240	-62.760	-62.857	40.00	1.03×10^8	98.3	<i>Local Stimulus Transient</i>
5	3.763	-61.237	-61.524	37.21	4.44×10^7	150.1	<i>Local Stimulus Transient</i>
10	4.929	-60.071	-60.612	34.78	2.27×10^7	209.7	<i>Local Stimulus Transient</i>
15	5.507	-59.493	-60.216	34.04	1.53×10^7	255.8	<i>Local Stimulus Transient</i>
20	5.829	-59.171	-60.036	32.65	1.15×10^7	294.8	<i>Local Stimulus Transient</i>
25	6.015	-58.985	-59.962	32.00	9.23×10^6	329.1	<i>Local Stimulus Transient</i>
30	6.121	-58.879	-59.950	30.19	7.71×10^6	360.3	<i>Local Stimulus Transient</i>
40	6.197	-58.803	-60.032	28.57	5.79×10^6	415.6	<i>Local Stimulus Transient</i>
50	6.184	-58.816	-60.186	27.12	4.64×10^6	464.3	<i>Local Stimulus Transient</i>

Table 5 Spatial grid refinement and convergence verification holding physical node width fixed ($l_{node} = 1.0 \mu\text{m}$, $\Delta t = 2.5 \mu\text{s}$, $w_{cleft} = 20 \text{ nm}$). A β excursion is measured at $z = 4 \text{ mm}$; “Converged” denotes a successive change below 0.5% in both ΔV columns.

Spatial Step Δz	Isolated A β CV (m/s)	Isolated A β Excursion (mV)	Single-Pair Δ V ($n=1$, mV)	Multi-Fiber Δ V ($n=25$, mV)	Classification	Status
20.0 μm	41.03	80.44	1.237	5.919	<i>Local Stimulus Transient</i>	Coarse reference
10.0 μm (Nominal)	41.03	80.44	1.325	6.015	<i>Local Stimulus Transient</i>	Nominal study grid
5.0 μm	41.03	80.44	1.347	6.041	<i>Local Stimulus Transient</i>	Converging
2.5 μm	41.03	80.44	1.352	6.047	<i>Local Stimulus Transient</i>	Converged ($<0.5\%$)
1.25 μm	41.03	80.44	1.353	6.049	<i>Local Stimulus Transient</i>	Converged ($<0.5\%$)
1.0 μm	41.03	80.44	1.354	6.050	<i>Local Stimulus Transient</i>	Converged ($<0.5\%$)

Declarations

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Ethics Approval

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Consent to Participate

Not applicable. No human participants were involved in this study.

Consent for Publication

Not applicable. This manuscript contains no individual person’s data in any form.

Data Availability

All numerical data supporting the findings of this study (raw simulation outputs, convergence studies, and reconciliation tables underlying every figure and table) are provided as Supplementary Information accompanying this article and are additionally archived at the code repository cited below. No restrictions apply to their reuse.

Code Availability

The simulation code implementing the closed-loop core-conductor model, CRRSS kinetics, Nav1.8/Nav1.9 nociceptor kinetics, the multi-fiber bundle formulation, and all verification suites is freely available for academic use at: <https://github.com/MinaSaied23/ephaptic-crosstalk>.

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Author Contributions

Mina Saied Attia Rizk: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing - Original Draft, Writing - Review & Editing, Visualization.

References

- Akopian, A. N., Sivilotti, L., & Wood, J. N. (1996). A tetrodotoxin-resistant voltage-gated sodium channel expressed by sensory neurons. *Nature*, 379(6562), 257–262. <https://doi.org/10.1038/379257a0>
- Amir, R., & Devor, M. (2000). Functional cross-excitation between afferent A- and C-neurons in dorsal root ganglia. *Neuroscience*, 95(1), 189–195. [https://doi.org/10.1016/s0306-4522\(99\)00418-4](https://doi.org/10.1016/s0306-4522(99)00418-4)
- Arvanitaki, A. (1942). Effects evoked in an axon by the activity of a contiguous one. *Journal of Neurophysiology*, 5(2), 89–108. <https://doi.org/10.1152/jn.1942.5.2.89>
- Asmedi, A., Wibowo, S., & Meliala, L. (2018). Ephaptic crosstalk in painful diabetic neuropathy: An electrodiagnostic study. *Journal of the Medical Sciences (Berkala Ilmu Kedokteran)*, 50(2), 173–179. <https://doi.org/10.19106/JMedSci005002201806>
- Chawla, A., & Morgera, S. D. (2024). On the geometry of interacting axon tracts in relation to their functional properties. *Communications in Mathematical Biology and Neuroscience*, 2024, Article 57. <https://doi.org/10.28919/cmbn/8509>
- Chawla, A., Morgera, S. D., & Snider, A. D. (2021). On axon interaction and its role in neurological networks. *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, 18(2), 790–796. <https://doi.org/10.1109/TCBB.2019.2941689>
- Chiu, S. Y., Ritchie, J. M., Rogart, R. B., & Stagg, D. (1979). A quantitative description of membrane currents in rabbit myelinated nerve. *The Journal of Physiology*, 292, 149–166. <https://doi.org/10.1113/jphysiol.1979.sp012843>
- Cummins, T. R., Dib-Hajj, S. D., Black, J. A., Akopian, A. N., Wood, J. N., & Waxman, S. G. (1999). A novel persistent tetrodotoxin-resistant sodium current in SNS-null and wild-type small primary sensory neurons. *Journal of Neuroscience*, 19(24), RC43. <https://doi.org/10.1523/JNEUROSCI.19-24-j0001.1999>
- Devor, M., & Wall, P. D. (1990). Cross-excitation in dorsal root ganglia of nerve-injured and intact rats. *Journal of Neurophysiology*, 64(6), 1733–1744. <https://doi.org/10.1152/jn.1990.64.6.1733>
- Goldwyn, J. H., & Rinzel, J. (2016). Neuronal coupling by endogenous electric fields: Cable theory and applications to coincidence detector neurons in the auditory brain stem. *Journal of Neurophysiology*, 115(4), 2033–2051. <https://doi.org/10.1152/jn.00780.2015>

- Hodgkin, A. L., & Huxley, A. F. (1952). A quantitative description of membrane current and its application to conduction and excitation in nerve. *The Journal of Physiology*, 117(4), 500–544. <https://doi.org/10.1113/jphysiol.1952.sp004764>
- Jæger, K. H., & Tveito, A. (2022). Deriving the bidomain model of cardiac electrophysiology from a cell-based model; properties and comparisons. *Frontiers in Physiology*, 12, 811029. <https://doi.org/10.3389/fphys.2021.811029>
- Jæger, K. H., & Tveito, A. (2026). Extracellular stimulation and ephaptic coupling of neurons in a fully coupled finite element-based Extracellular-Membrane-Intracellular (EMI) model. *Frontiers in Computational Neuroscience*, 20, 1755548. <https://doi.org/10.3389/fncom.2026.1755548>
- Katz, B., & Schmitt, O. H. (1940). Electric interaction between two adjacent nerve fibres. *The Journal of Physiology*, 97(4), 471–488. <https://doi.org/10.1113/jphysiol.1940.sp003823>
- Klein, A. H., Vyshnevskaya, A., Hartke, T. V., De Col, R., Mankowski, J. L., Turnquist, B., Bosmans, F., Reeh, P. W., Schmelz, M., Carr, R. W., & Ringkamp, M. (2017). Sodium channel Nav1.8 underlies TTX-resistant axonal action potential conduction in somatosensory C-fibers of distal cutaneous nerves. *Journal of Neuroscience*, 37(20), 5204–5214. <https://doi.org/10.1523/JNEUROSCI.3799-16.2017>
- Lindén, H., Hagen, T., Łęski, S., Norheim, E. S., Pettersen, K. H., & Einevoll, G. T. (2014). LFPy: A tool for biophysical simulation of extracellular potentials generated by detailed model neurons. *Frontiers in Neuroinformatics*, 7, 41. <https://doi.org/10.3389/fninf.2013.00041>
- Murinson, B. B., & Griffin, J. W. (2004). C-fiber structure varies with location in peripheral nerve. *Journal of Neuropathology and Experimental Neurology*, 63(3), 246–254. <https://doi.org/10.1093/jnen/63.3.246>
- Pelot, N. A., Catherall, D. C., Thio, B. J., Titus, N. D., Liang, E. D., Henriquez, C. S., & Grill, W. M. (2021). Excitation properties of computational models of unmyelinated peripheral axons. *Journal of Neurophysiology*, 125(1), 86–104. <https://doi.org/10.1152/jn.00315.2020>
- Rappaport, Z. H., & Devor, M. (1994). Trigeminal neuralgia: The role of self-sustaining discharge (ignition) in the medicine and surgery of paroxysmal pain. *Pain*, 56(2), 127–138. [https://doi.org/10.1016/0304-3959\(94\)90086-6](https://doi.org/10.1016/0304-3959(94)90086-6)
- Rasminsky, M. (1980). Ephaptic transmission between single nerve fibres in the spinal nerve roots of dystrophic mice. *The Journal of Physiology*, 305(1), 151–169. <https://doi.org/10.1113/jphysiol.1980.sp013356>
- Reutskiy, S., Rossoni, E., & Tirozzi, B. (2003). Conduction in bundles of demyelinated nerve fibers: Computer simulation. *Biological Cybernetics*, 89(6), 439–448. <https://doi.org/10.1007/s00422-003-0430-x>
- Schmidt, H., Hahn, G., Deco, G., & Knösche, T. R. (2021). Ephaptic coupling in white matter fibre bundles modulates axonal transmission delays. *PLOS Computational Biology*, 17(2), e1007858. <https://doi.org/10.1371/journal.pcbi.1007858>
- Schmidt, H., & Knösche, T. R. (2022). Modelling the effect of ephaptic coupling on spike propagation in peripheral nerve fibres. *Biological Cybernetics*, 116(4), 461–473. <https://doi.org/10.1007/s00422-022-00934-9>

- Seltzer, Z., & Devor, M. (1979). Ephaptic transmission in chronically damaged peripheral nerves. *Neurology*, 29(7), 1061–1064. <https://doi.org/10.1212/wnl.29.7.1061>
- Sundt, D., Gamper, N., & Jaffe, D. B. (2015). Spike propagation through the dorsal root ganglia in an unmyelinated sensory neuron: A modeling study. *Journal of Neurophysiology*, 114(6), 3140–3153. <https://doi.org/10.1152/jn.00226.2015>
- Sweeney, J. D., Mortimer, J. T., & Durand, D. M. (1987). Modeling of mammalian myelinated nerve for functional neuromuscular stimulation. In *Proceedings of the 9th Annual Conference of the IEEE Engineering in Medicine and Biology Society* (pp. 1577–1578).
- Tigerholm, J., Petersson, M. E., Obreja, O., Lampert, A., Carr, R., Schmelz, M., & Fransén, E. (2014). Modeling activity-dependent changes of axonal spike conduction in primary afferent C-nociceptors. *Journal of Neurophysiology*, 111(9), 1721–1735. <https://doi.org/10.1152/jn.00777.2012>